

Battery Fast Charging Optimization

**Revised Simscape model, reproducible studies
and verification**

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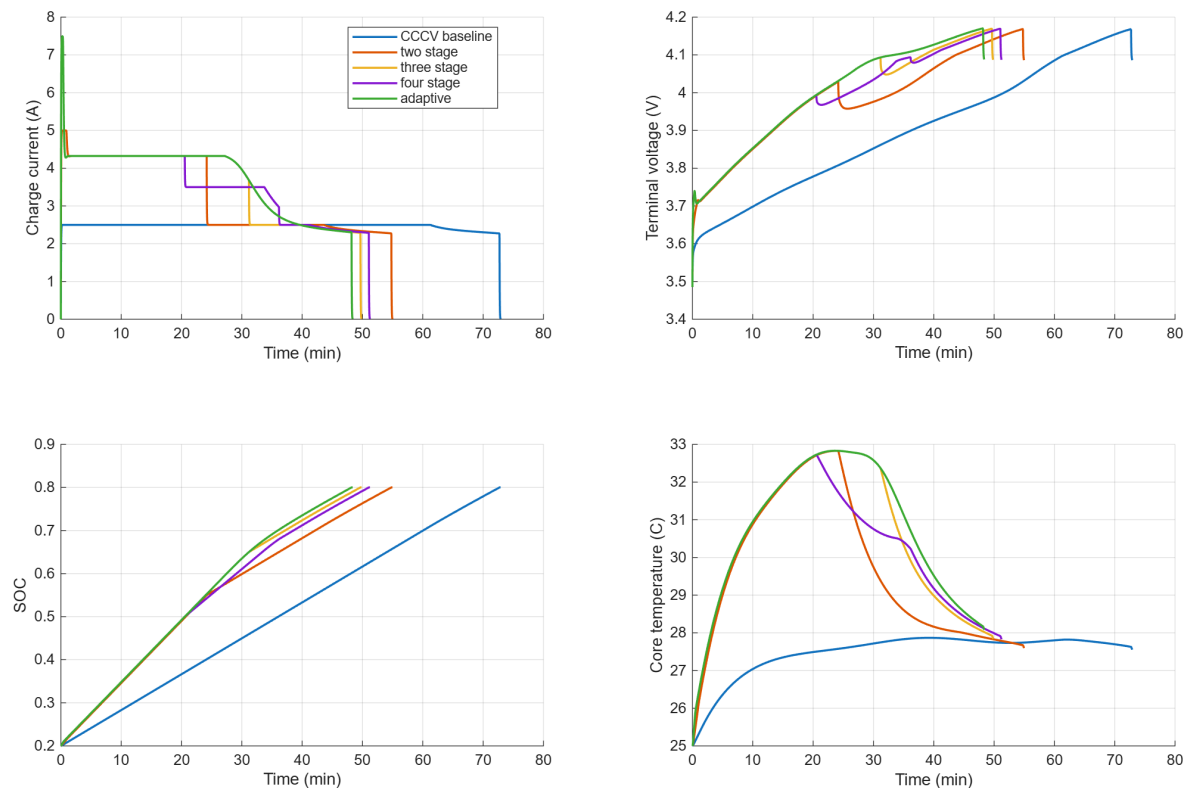
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Chapter 1. Executed results and recommendation

The baseline takes 72.71 minutes; the adaptive request takes 48.18 minutes; the best tested staged request takes 48.18 minutes. The adaptive reduction relative to the baseline is 33.7%. These are newly executed simulation results under a shared protection policy.

The staged-to-adaptive time difference is 0.00 seconds. Interpret a tie as evidence that the common derating supervisor dominates the request. It is not evidence of an optimizer advantage. Binding shares are elapsed-time weighted and exclude the shutdown tail.

name	minutes	max voltage V	max core C	feasible
CCCV baseline	72.71	4.169	27.87	1
two_stage	54.78	4.169	32.83	1
three_stage	49.66	4.17	32.83	1
four_stage	51.05	4.17	32.73	1
adaptive	48.18	4.171	32.83	1



Chapter 2. Model and protection architecture

Executed regression checks: passed = 1. The forced-voltage-fault trace ends at 0.00868 A after gated shutdown. Enabled SEI starts at 5e-09 m. Unequal electrolyte layer counts, inactive-aging telemetry, live limit parameters and the shared external clamp were exercised.

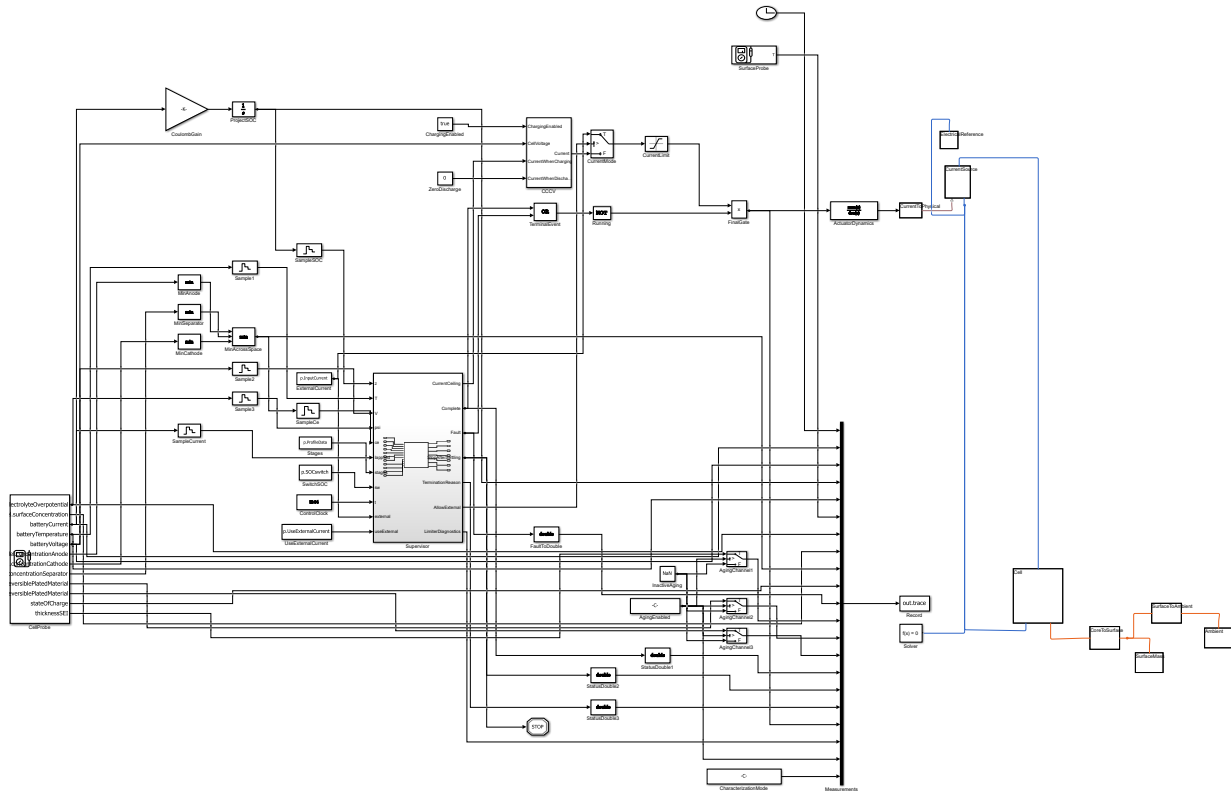
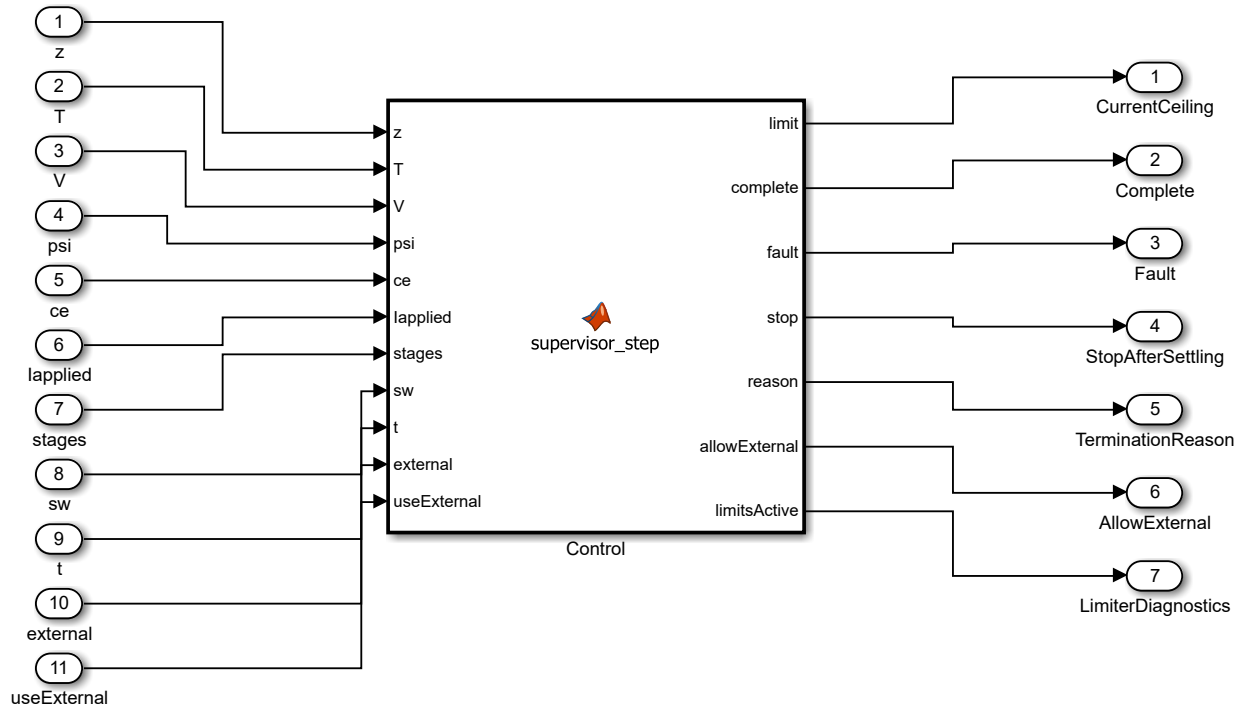


Figure 2.1. Generated protected Simscape plant; inspect the source model for block dialogs.



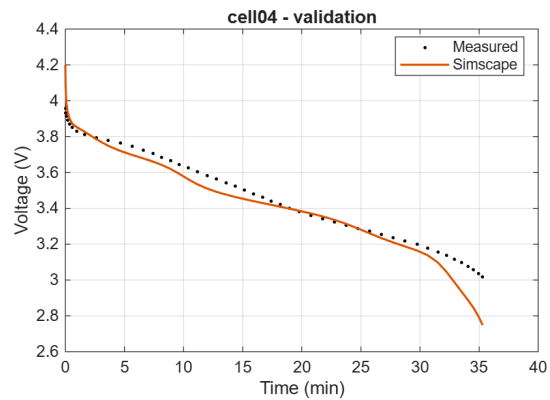
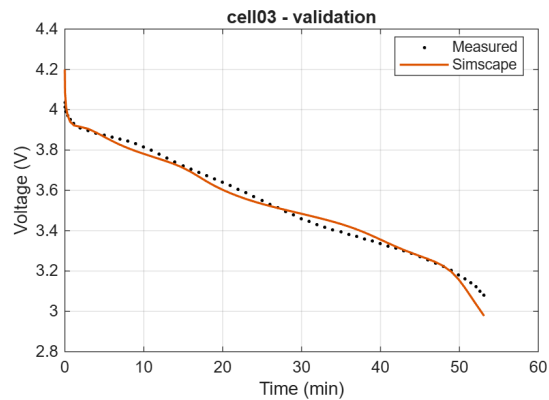
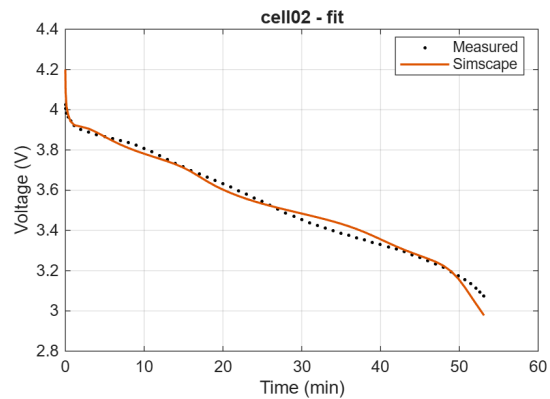
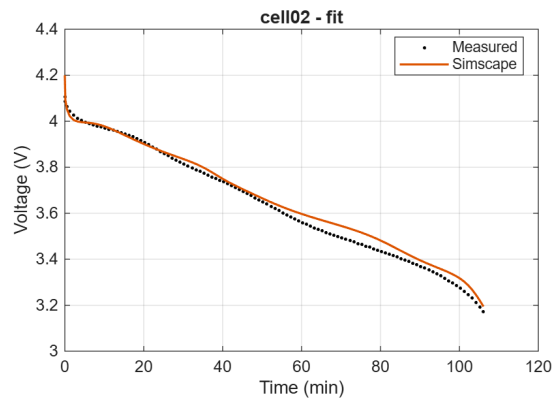
Chapter 3. Experiments and native battery diagnostics

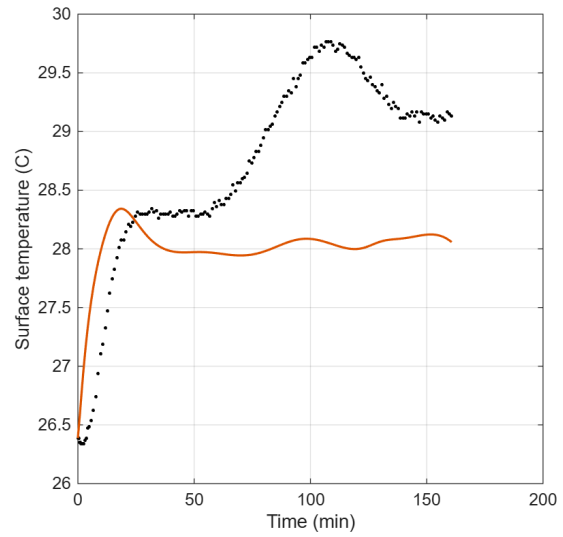
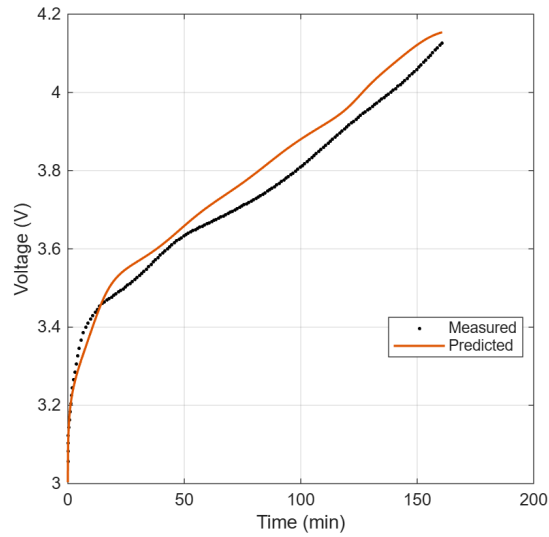
cell	role	current A	voltage RMSE after mV	temperature RMSE C	coverage fraction
cell02	fit	2.5	30.65	0.7568	1
cell02	fit	5	39.85	1.778	1
cell03	validation	5	39.79	2.273	1
cell04	validation	7.5	91.47	2.838	1

cell03 held-out 5.00 A discharge: voltage RMSE changes from 54.26 to 39.79 mV. Temperature RMSE is 2.27 C. The fit is not assumed to improve every held-out rate.

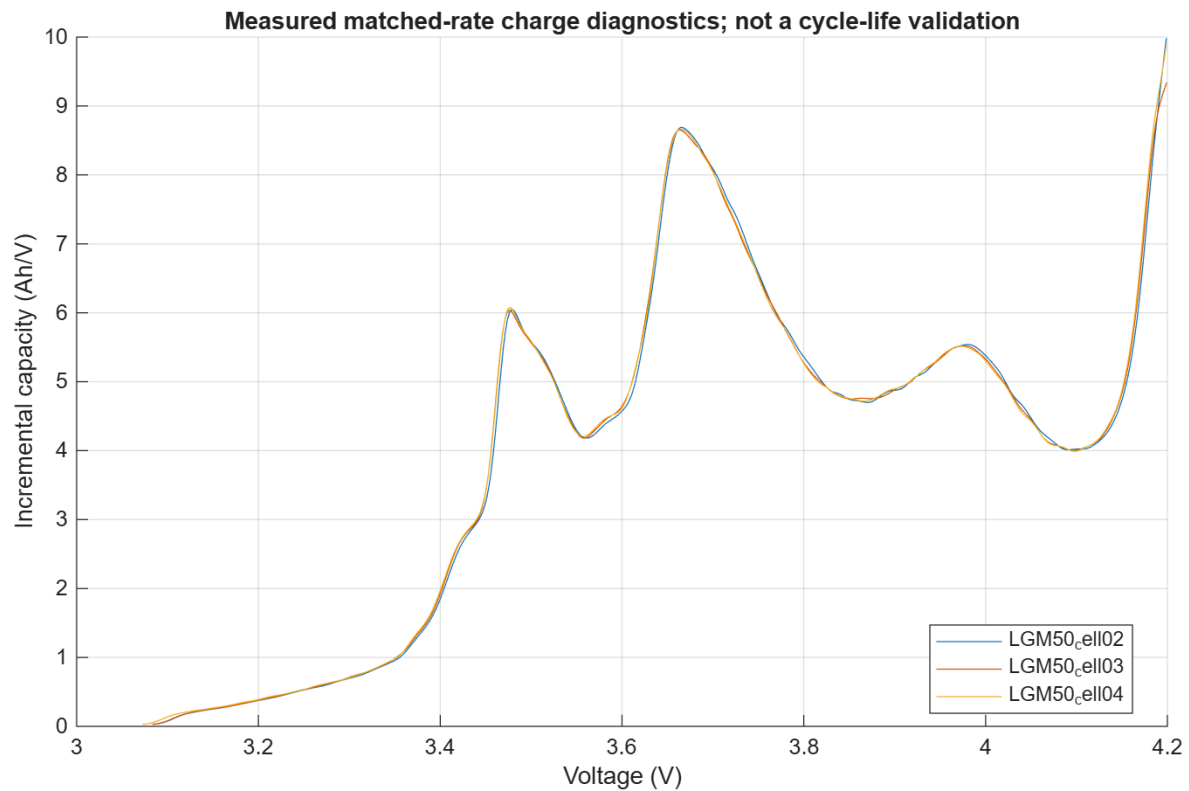
cell04 held-out 7.50 A discharge: voltage RMSE changes from 55.25 to 91.47 mV. Temperature RMSE is 2.84 C. The fit is not assumed to improve every held-out rate.

In particular, deterioration on a higher-rate holdout limits transfer of the fitted parameters to fast charging. Full window coverage is a reproducibility check, not an accuracy acceptance criterion. High-rate charging remains experimentally unvalidated.





cell	peak	peakLocation	area
LGM50_cell02	8.689	3.666	4.447
LGM50_cell03	8.654	3.664	4.469
LGM50_cell04	8.665	3.665	4.476

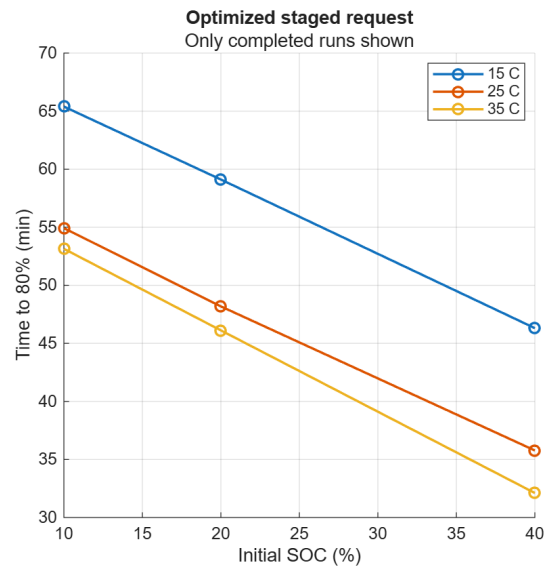
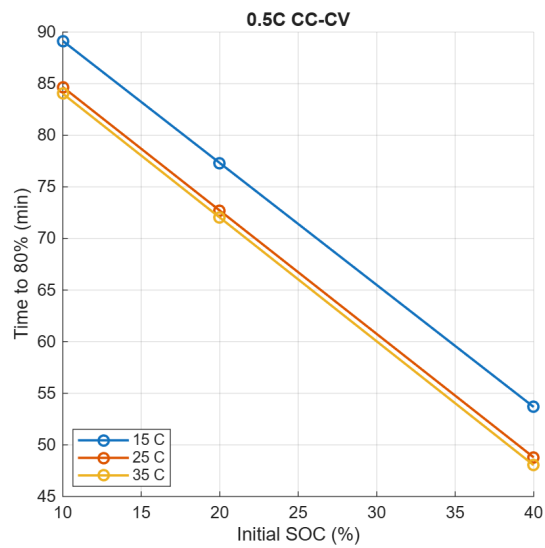


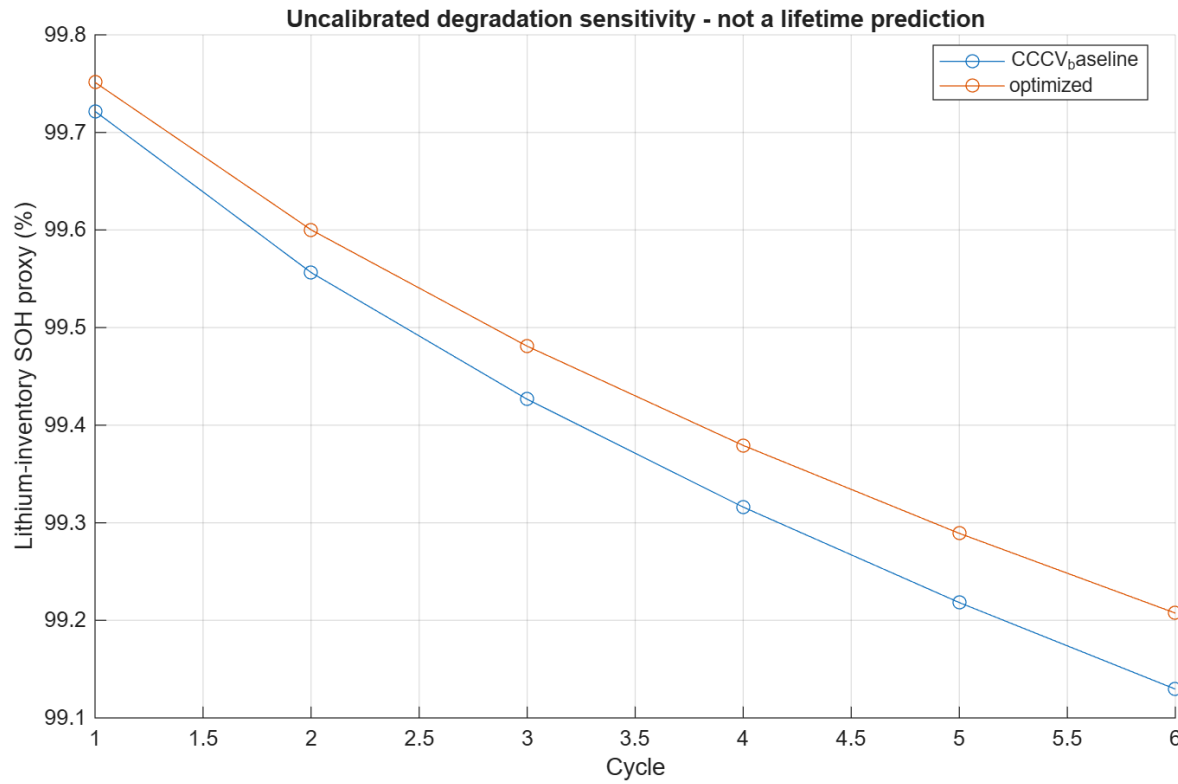
Matched low-rate IC curves compare three characterized cells. Peak differences are not a measured aging trajectory and do not validate the simulated lithium-inventory proxy.

Chapter 4. Operating grid, numerical checks and aging

18 of 18 operating-grid cases meet the configured charging feasibility checks. Failed cases remain visible in the result table and cannot be counted as completed charges.

temperature C	initial soc	profile	minutes	termination reason	feasible
15	0.1	1	89.14	1	1
15	0.1	2	65.39	1	1
15	0.2	1	77.32	1	1
15	0.2	2	59.11	1	1
15	0.4	1	53.67	1	1
15	0.4	2	46.31	1	1
25	0.1	1	84.67	1	1
25	0.1	2	54.92	1	1
25	0.2	1	72.71	1	1
25	0.2	2	48.18	1	1
25	0.4	1	48.79	1	1
25	0.4	2	35.77	1	1
35	0.1	1	84.03	1	1
35	0.1	2	53.14	1	1
35	0.2	1	72.03	1	1
35	0.2	2	46.12	1	1
35	0.4	1	48.03	1	1
35	0.4	2	32.11	1	1

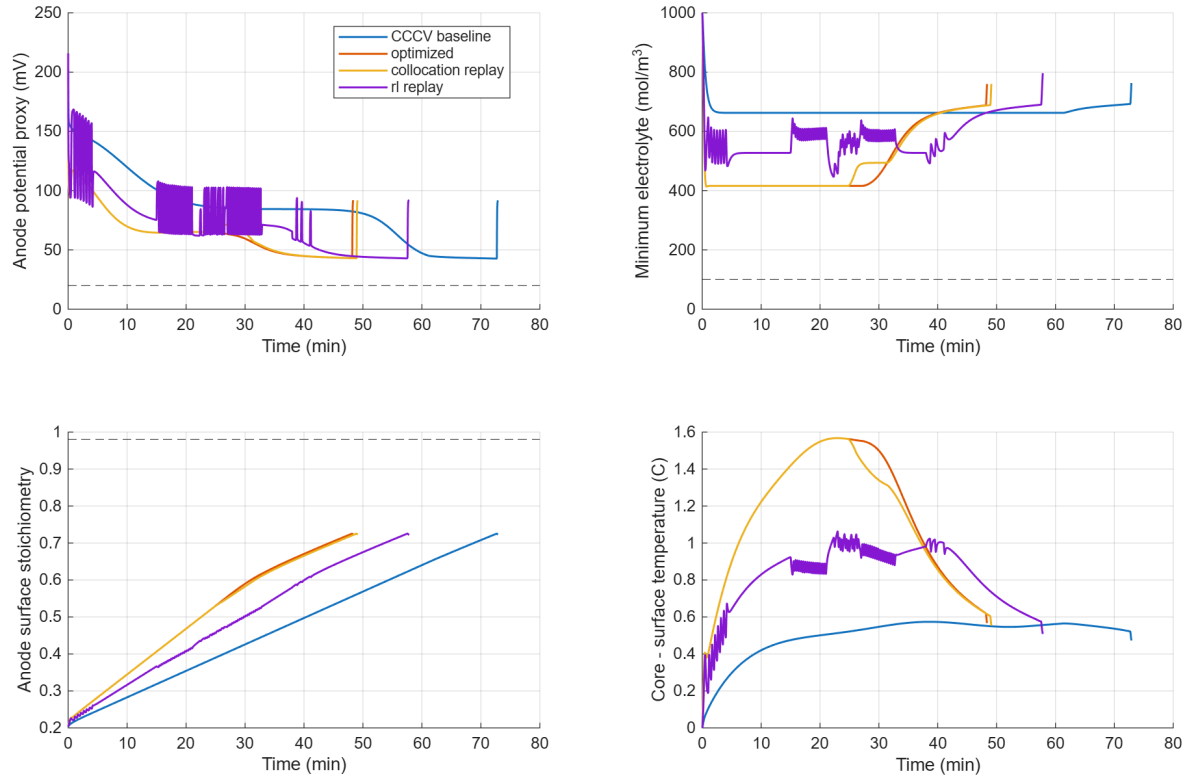




The six-cycle aging comparison is an uncalibrated model sensitivity. Enabled degradation is reproducible; neither this table nor IC features establish a lifetime improvement.

Chapter 5. Advanced algorithms and reproducibility

Reinforcement Learning Toolbox trained 100 episodes directly on the full plant using seed 42. Greedy evaluation took 57.57 minutes; feasible = 1. Training statistics and the agent are saved in results/learned_policy.mat.



step	executed	wall seconds
model_build	1	5.183
protection_regressions	1	25.18
native_battery_diagnostics	1	8.694
charging_comparison	1	70.47
simscape_optimization	1	186.5
SDO_experimental_validation	1	818.1
direct_collocation	1	2.071
RL_full_plant	1	1386
parallel_robustness_and_refinement	1	105
aging	1	53.45
charge_validation_and_sensitivity	1	29.69
final_checks	1	8.657
native_report	1	15
recorded_trace_demo	1	199

Read verification/final_checks.json and revision_regressions.json for assertions. A pipeline stage is executed only after returning successfully. Optional MPC is not implemented. Historical outputs are excluded from this report.

Chapter 6. Methods, scope and references

Scope and data

This study compares requests for charging a nominal 5 Ah LG M50 cell from 20% to 80% coulomb-counted SOC. The published Chen et al. parameterization fixes geometry, stoichiometry and OCP curves. The redistributed raw records are the authors' public cell02, cell03 and cell04 characterization files (Zenodo 4032561, CC BY 4.0). They contain low-rate charging and several discharge rates, not high-rate charge cycle-life validation.

The proposed currents, thermal fixture and constraint values are study assumptions. The model is not evidence of manufacturer-approved charging or of experimentally validated plating prevention. Its anode potential and lithium-inventory loss are model outputs, not measurements. The separate historical paper and video predate this revision and cannot substantiate its numerical results.

Electrochemical and thermal model

The Simscape Battery Single Particle block includes solid diffusion and electrolyte dynamics. Each electrode uses 12 radial shells; the three electrolyte regions default to 12 layers each and may now be configured independently. The spatial minimum is computed within each region before comparing the three scalar minima. The OCP lookup has 600 absolute-stoichiometry points.

Positive current charges the cell. Reported SOC follows $dz/dt = I/(3600 Q)$, with $Q = 5$ Ah; the block's internal normalized electrode SOC is also logged. Core thermal capacity is 42.775 J/K and surface capacity is 8 J/K. Explicit Foundation Thermal Resistance blocks connect core to surface (1.5 K/W) and surface to ambient (6 K/W). These thermal assumptions are distinct from the effective resistance fitted to experiments.

Controller and termination

The four stage requests are derived from p.Istages at run time. A MATLAB Function supervisor receives named measurements and sets the charge-current ceiling to the minimum of the request, the 7.5 A ceiling and three proportional limits: thermal, anode potential and electrolyte concentration. The electrolyte term is $\max(0, 0.02 (ce_min - 100 - 100))$ A. The additional 100 mol/m³ soft margin leaves room for diffusion and actuator dynamics before the hard 100 mol/m³ floor. All gains, soft margins and hard thresholds are named parameters.

The built-in Battery CC-CV controller regulates to 4.195 V under a 4.2 V hard ceiling. A shared downstream saturation clamps both operating modes to [-7.5, 7.5] A. A shared terminal gate commands zero on completion or fault, followed by a two-second actuator time constant. The model stops when measured current is at most 0.01 A after at least

0.5 seconds of settling, or after a 20-second maximum shutdown window. The recorded terminal current makes the distinction inspectable.

Termination codes are 0 running, 1 target completed, 2 high voltage, 3 high temperature, 4 low anode potential, 5 electrolyte depletion, 6 low voltage/SOC, 7 nonfinite measurement, 8 duration expired without completion, and 9 scheduled diagnostic replay finished. Codes 8 and 9 are assigned by postprocessing. A fault never qualifies as a completed charge. Duration is time to the completion event when reached; `stop_time_s` includes shutdown settling. Delivered charge and energy include the tail.

External replay and degradation modes

Normal positive external requests pass through the same derating and CC-CV regulator. Negative current uses the shared bipolar clamp and hard protection. `CharacterizationReplay` is an explicit opt-in for matching measured current or charge-balanced aging schedules: it bypasses charging-specific soft limits and CV regulation, but retains the downstream current clamp, hard diagnostic guards and shutdown gate. Its declared diagnostic limits are 4.25 V, 2.0 V, 60 C, 1 mol/m³ electrolyte and -50 mV anode potential. Such traces are never classified as feasible charging policies.

`EnableAging` selects diffusion-limited SEI growth and fractional reversible plating through `SimulationInput` block overrides. Enabled SEI starts at 5 nm and is asserted against the logged state. Disabled degradation outputs are NaN with `aging_enabled = 0`. The six-cycle demonstration uses illustrative side-reaction parameters, replays charge-balanced schedules and computes a lithium-inventory SOH proxy. It does not predict measured capacity retention or lifetime. Measured incremental-capacity curves provide a separate cross-cell diagnostic, not a calibration of these side reactions.

Optimization, fitting and reinforcement learning

Pattern search evaluates staged requests directly against the protected Simscape model. Its finite budget supports a best-tested candidate claim, not a global optimum. Trapezoidal direct collocation remains explicitly a reduced-model calculation; its requested current must then complete a full-plant replay. A lower surrogate time alone is not accepted as a charging improvement.

Grouped estimation uses Simulink Design Optimization `SimulationTest` objects, `SignalTracking` residual requirements and `sdo.optimize`. Two cell02 discharge records provide the objective; cell03 and cell04 discharge windows and cell04 low-rate charging are held out. Effective anode/cathode diffusivities and surface-to-ambient resistance are estimated with geometry and OCP fixed. Each record uses its first 90% duration and every second sample. Voltage residuals are scaled by 30 mV and temperature residuals by 1 C, then by the square root of each record's sample count. Full-window coverage is required. The dataset contains no EIS spectra, so no measured impedance fit is claimed.

Reinforcement Learning Toolbox Q-learning trains directly through `rlSimulinkEnv` on the same protected electrochemical plant. The tabular observation encodes 16 SOC bins, 8 temperature bins and 8 electrolyte-concentration bins. Five actions request 1.5, 2.5, 3.5,

5 or 7.5 A every 10 seconds. The reward combines SOC progress, elapsed-time cost, a warm-cell penalty and terminal reward/penalty. Seed 42, training options, statistics and agent are saved. The nominal comparison uses a greedy full-plant evaluation with the shutdown tail. Finite training does not establish convergence or superiority.

Reproducibility and interpretation

Profile, operating-condition and mesh sweeps use independent SimulationInput objects and parsim. The model stores no author-machine root path. Fresh-clone execution starts with start_project.m and run_project; the latter records each stage and fails explicitly if a required stage fails. Native MATLAB/Simulink Report Generator produces this PDF. Python is optional for the separate PyBaMM cross-check, whose electrolyte transport laws and CV implementation differ from Simscape.

The binding limiter is logged and its fraction is weighted by elapsed simulation time. Similar staged and adaptive results can arise when the common safety supervisor dominates the requested schedule. Report this tie prominently rather than attributing it to a sophisticated optimizer. The operating grid covers exact initial conditions only; it does not prove a continuous safe operating envelope. Optional MPC remains future work.

Primary references

Chen C-H et al. Development of Experimental Techniques for Parameterization of Multi-scale Lithium-ion Battery Models. Journal of The Electrochemical Society 167 (2020), 080534. DOI: 10.1149/1945-7111/ab9050.

Chen C-H et al. Experimental dataset. Zenodo record 4032561. <https://zenodo.org/records/4032561> (CC BY 4.0; full attribution in data/DATA_ATTRIBUTION.md).

MathWorks. Battery Single Particle; Battery CC-CV. <https://www.mathworks.com/help/simscape-battery/ref/batterysingleparticle.html> and <https://www.mathworks.com/help/simscape-battery/ref/batterycccv.html>.

MathWorks. sdo.SimulationTest; sdo.requirements.SignalTracking; sdo.optimize. <https://www.mathworks.com/help/slido/ref/sdo.simulationtest.html>.

MathWorks. rlSimulinkEnv and rlTable. <https://www.mathworks.com/help/reinforcement-learning/ref/rlsimulinkenv.html> and <https://www.mathworks.com/help/reinforcement-learning/ref/rltable.html>.

MathWorks. batteryTestDataParser; batteryTestFeatureExtractor; batteryDifferentialCurves; batteryDifferentialCurveFeatures. <https://www.mathworks.com/help/predmaint/ref/batterytestdataparser.html>.